

Quadrupole Radiation and Orbital Decay in Binary Pulsars

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Context

The conceptual structure of this derivation follows material introduced in advanced coursework at the Universitat de Barcelona. All detailed calculations, formal development, explicit algebraic expansions, and typesetting are the independent work of the author.

Introduction

We study the gravitational quadrupole radiation emitted by a binary system comprising two neutron stars, A and B , of identical mass M , orbiting their common center of mass in a perfectly circular orbit of radius R . The total radiated power dE/dt across all solid angles is determined by the third temporal derivative of the reduced quadrupole moment $\mathcal{P}^{ij}$:

$$\frac{dE}{dt} = \frac{G_N}{5c^5} \langle \ddot{\mathcal{P}}^{ij} \ddot{\mathcal{P}}^{ij} \rangle$$

where $\mathcal{P}^{ij} = P^{ij} - \frac{1}{3}\delta^{ij}P_k^k$, and $P^{ij} = \int \rho(\mathbf{r}, t) r^i r^j d^3r$ is the standard quadrupole moment.

1. Birkhoff's Theorem and the Reduced Quadrupole Moment

The necessity of utilizing the trace-free reduced quadrupole moment $\mathcal{P}^{ij}$ becomes evident when considering spherically symmetric mass distributions. For a density profile dependent strictly on the radial distance, $\rho(\mathbf{r}, t) = \rho(r, t)$, the standard quadrupole moment trace is $P_k^k = \int \rho(\mathbf{r}, t) r^k r_k d^3r = \int \rho(r, t) r^2 d^3r$. We evaluate $\mathcal{P}^{ij}$ by decomposing the position vector $r^i = n^i r$, where n^i is the unit normal, and applying the solid angle integral $\int_{\Omega} n^i n^j d\Omega = \frac{4\pi}{3}\delta^{ij}$:

$$\begin{aligned} \mathcal{P}^{ij} &= \int \left(r^i r^j - \frac{1}{3}\delta^{ij} r^2 \right) \rho(r, t) d^3r = \int \left(n^i n^j r^2 - \frac{1}{3}\delta^{ij} r^2 \right) \rho(r, t) r^2 dr d\Omega \\ &= \int r^4 \rho(r, t) dr \int_{\Omega} \left(n^i n^j - \frac{1}{3}\delta^{ij} \right) d\Omega = \int r^4 \rho(r, t) dr \left(\frac{4\pi}{3}\delta^{ij} - \frac{4\pi}{3}\delta^{ij} \right) = 0 \end{aligned}$$

This rigorously satisfies Birkhoff's theorem: a spherically symmetric mass distribution cannot emit gravitational radiation.

2. Quadrupole Moment of the Binary System

For a system of s discrete point masses identical in mass M located at $\mathbf{r}_s$, the mass density is $\rho(\mathbf{r}, t) = M \sum_s \delta^3(\mathbf{r} - \mathbf{r}_s)$. The standard quadrupole moment evaluates to:

$$P^{ij} = \int \rho(\mathbf{r}, t) r^i r^j d^3r = \int M \sum_s \delta^3(\mathbf{r} - \mathbf{r}_s) r_s^i r_s^j d^3r = M \sum_s \int r_s^i r_s^j \delta^3(\mathbf{r} - \mathbf{r}_s) d^3r = M \sum_s r_s^i r_s^j$$

Choosing a coordinate system centered on the center of mass such that the two identical masses A and B orbit symmetrically, we have $\mathbf{r}_A = -\mathbf{r}_B = \mathbf{r}$. Substituting this yields:

$$P^{ij} = Mr^i r^j + M(-r^i)(-r^j) = 2Mr^i r^j$$

3. Temporal Derivatives and the Circular Orbit Condition

We compute the successive temporal derivatives of the standard quadrupole moment tensor by explicitly applying the product rule:

$$\dot{P}^{ij} = 2M (\dot{r}^i r^j + r^i \dot{r}^j)$$

$$\ddot{P}^{ij} = 2M (\ddot{r}^i r^j + \dot{r}^i \dot{r}^j + \dot{r}^i \dot{r}^j + r^i \ddot{r}^j) = 2M (\ddot{r}^i r^j + 2\dot{r}^i \dot{r}^j + r^i \ddot{r}^j)$$

$$\dddot{P}^{ij} = 2M (\dddot{r}^i r^j + \ddot{r}^i \dot{r}^j + 2\dot{r}^i \ddot{r}^j + 2\dot{r}^i \ddot{r}^j + \dot{r}^i \ddot{r}^j + r^i \dddot{r}^j) = 2M (\dddot{r}^i r^j + 3\ddot{r}^i \dot{r}^j + 3\dot{r}^i \ddot{r}^j + r^i \dddot{r}^j)$$

For a strictly circular orbit with constant velocity magnitude v , the centripetal acceleration dictates $\ddot{r}^i = a^i = -\frac{v^2}{R^2} r^i$, which implies $\ddot{r}^i = -\frac{v^2}{R^2} r^i$. We substitute these dynamic constraints into the third derivative:

$$\begin{aligned} \dddot{P}^{ij} &= 2M \left(-\frac{v^2}{R^2} \dot{r}^i r^j - 3\frac{v^2}{R^2} r^i \dot{r}^j - 3\dot{r}^i \frac{v^2}{R^2} r^j - r^i \frac{v^2}{R^2} \dot{r}^j \right) \\ &= -\frac{2Mv^2}{R^2} (\dot{r}^i r^j + 3r^i \dot{r}^j + 3\dot{r}^i r^j + r^i \dot{r}^j) = -\frac{8Mv^2}{R^2} (\dot{r}^i r^j + r^i \dot{r}^j) \end{aligned}$$

Since $\mathbf{r}$ and $\mathbf{v} = \dot{\mathbf{r}}$ are orthogonal for circular orbits ($\mathbf{r} \cdot \dot{\mathbf{r}} = 0$), the trace is $\ddot{P}_k^k = 0$. Because the trace of the third derivative vanishes, the reduced quadrupole derivative perfectly coincides with the standard quadrupole derivative: $\ddot{P}^{ij} = \ddot{P}^{ij}$.

4. Explicit Tensor Components

We constrain the orbital plane to XY with angular velocity ω . The positions and velocities are:

$$\begin{aligned} r^x &= R \cos(\omega t), & r^y &= R \sin(\omega t), & r^z &= 0 \\ \dot{r}^x &= -R\omega \sin(\omega t), & \dot{r}^y &= R\omega \cos(\omega t), & \dot{r}^z &= 0 \end{aligned}$$

Inserting these kinematics isolates the non-vanishing tensor components:

$$\begin{aligned} \ddot{P}^{xx} &= -\frac{8Mv^2}{R^2} (\dot{r}^x r^x + r^x \dot{r}^x) = -\frac{16Mv^2}{R^2} r^x \dot{r}^x = +\frac{16Mv^2}{R^2} R\omega \sin(\omega t) R \cos(\omega t) \\ &= 8Mv^2 \omega \sin(2\omega t) \\ \ddot{P}^{yy} &= -\frac{8Mv^2}{R^2} (\dot{r}^y r^y + r^y \dot{r}^y) = -\frac{16Mv^2}{R^2} r^y \dot{r}^y = -\frac{16Mv^2}{R^2} R\omega \sin(\omega t) R \cos(\omega t) \\ &= -8Mv^2 \omega \sin(2\omega t) \\ \ddot{P}^{yx} &= \ddot{P}^{xy} = -\frac{8Mv^2}{R^2} (\dot{r}^x r^y + r^x \dot{r}^y) = -\frac{8Mv^2}{R^2} (-R^2 \omega \sin^2(\omega t) + R^2 \omega \cos^2(\omega t)) \end{aligned}$$

$$= 8Mv^2 \left(\sin^2(\omega t) - \cos^2(\omega t) \right) = -8Mv^2 \omega \cos(2\omega t)$$

5. Total Radiated Power and Dimensional Analysis

To determine the radiated power, we compute the full contraction of the tensor:

$$\begin{aligned} \ddot{P}^{ij} \ddot{P}^{ij} &= \ddot{P}^{xx} \ddot{P}^{xx} + 2\ddot{P}^{xy} \ddot{P}^{xy} + \ddot{P}^{yy} \ddot{P}^{yy} \\ &= \left(8Mv^2 \omega \sin(2\omega t) \right)^2 + \left(-8Mv^2 \omega \sin(2\omega t) \right)^2 + 2 \left(-8Mv^2 \omega \cos(2\omega t) \right)^2 \\ &= 2 \left(8Mv^2 \omega \sin(2\omega t) \right)^2 + 2 \left(8Mv^2 \omega \cos(2\omega t) \right)^2 = 2 \left(8Mv^2 \omega \right)^2 = 128M^2 v^4 \omega^2 \end{aligned}$$

Taking the temporal average and substituting $v = R\omega$:

$$\frac{dE}{dt} = \frac{G_N}{5c^5} 128M^2 v^4 \omega^2 = \frac{128G_N M^2 R^4 \omega^6}{5c^5}$$

The orbital angular velocity is found via Newtonian mechanics: $\omega = \sqrt{\frac{G_N M}{4R^3}}$. Substituting this yields the final power:

$$\frac{dE}{dt} = \frac{128G_N M^2 R^4}{5c^5} \left(\frac{G_N M}{4R^3} \right)^3 = \frac{128G_N M^2 R^4}{5c^5} \frac{G_N^3 M^3}{64R^9} = \frac{2}{5} \frac{G_N^4 M^5}{R^5 c^5}$$

A dimensional analysis in natural units ($c = 1 \implies [M] = [E] = L^{-1}, [R] = [t] = L$) verifies the consistency. In SI, $[G_N] = \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, so naturally $[G_N] = L^3 L L^{-2} = L^2$:

$$\left[\frac{dE}{dt} \right] = \frac{E}{L} = L^{-2} \quad \text{and} \quad \left[\frac{2}{5} \frac{G_N^4 M^5}{R^5 c^5} \right] = \frac{L^8 L^{-5}}{L^5} = L^{-2}$$

6. Orbital Decay of the Binary Pulsar

The binary system progressively loses total orbital energy E , causing a decay in the rotational period T . Calculating the total mechanical energy $E = K_A + K_B + U$:

$$\begin{aligned} K_A &= K_B = \frac{1}{2} M v^2 = \frac{1}{2} M (R\omega)^2 = \frac{1}{2} M R^2 \left(\frac{G_N M}{4R^3} \right) = \frac{G_N M^2}{8R} \\ U &= -\frac{G_N M^2}{2(2R)} = -\frac{G_N M^2}{4R} \implies E = 2 \left(\frac{G_N M^2}{8R} \right) - \frac{G_N M^2}{4R} = -\frac{G_N M^2}{4R} \end{aligned}$$

We eliminate R in favor of T using $\omega = 2\pi/T$:

$$T = 2\pi \sqrt{\frac{4R^3}{G_N M}} = \sqrt{\frac{16\pi^2 R^3}{G_N M}} \implies R^3 = \frac{G_N M T^2}{16\pi^2} \implies R = \left(\frac{G_N M T^2}{16\pi^2} \right)^{1/3}$$

Substituting this radius $R(T)$ back into the energy equation:

$$E = -\frac{G_N M^2}{4 \left(\frac{G_N M T^2}{16\pi^2} \right)^{1/3}} = -\left(\frac{16\pi^2 G_N^3 M^6}{4^3 G_N M T^2} \right)^{\frac{1}{3}} = -\left(\frac{\pi^2 G_N^2 M^5}{4 T^2} \right)^{\frac{1}{3}} = -M \left(\frac{\pi G_N M}{2 T} \right)^{\frac{2}{3}}$$

To isolate the time derivative, we define a constant $A := \left(\frac{1}{2}\pi G_N M\right)^{\frac{2}{3}}$ and differentiate:

$$\begin{aligned} E = -MAT^{-\frac{2}{3}} &\implies \frac{dE}{dt} = \frac{2}{3}MAT^{-\frac{2}{3}}T^{-1}\frac{dT}{dt} = -\frac{2}{3}\left(-MAT^{-\frac{2}{3}}\right)T^{-1}\frac{dT}{dt} = -\frac{2}{3}\frac{E}{T}\frac{dT}{dt} \\ &\implies \dot{T} = -\frac{3}{2}\frac{T}{E}\frac{dE}{dt} \end{aligned}$$

Substituting the full expressions for E and dE/dt :

$$\begin{aligned} \frac{dT}{dt} &= -\frac{3}{2}\frac{T}{-M\left(\frac{\pi G_N M}{2T}\right)^{\frac{2}{3}}}\left(\frac{2}{5}\frac{G_N^4 M^5}{c^5}\left(\frac{16\pi^2}{G_N M T^2}\right)^{\frac{5}{3}}\right) \\ &= -\frac{3}{5}\frac{1}{M}\left(\frac{2^2}{\pi^2 G_N^2 M^2}\frac{16^5 \pi^{10}}{G_N^5 M^5 T^5}\right)^{\frac{1}{3}}\frac{G_N^4 M^5}{c^5} = -\frac{3}{5c^5}\left(\frac{4 \cdot 16^5 \pi^8}{G_N^7 M^7 T^5}G_N^{12}M^{12}\right)^{\frac{1}{3}} \\ &= -\frac{3 \cdot 4^2}{5c^5}\left(\frac{4^5 \pi^5 G_N^5 M^5}{c^5}\right)^{\frac{1}{3}} = -\frac{48\pi}{5c^5}\left(\frac{4\pi G_N M}{T}\right)^{\frac{5}{3}} \end{aligned}$$

Hulse-Taylor Pulsar Orbital Decay

For elliptical orbits with eccentricities $e \neq 0$, exact General Relativistic mechanics introduces a correction factor $K_e = (1 + \frac{73}{24}e^2 + \frac{37}{96}e^4)/(1 - e^2)^{7/2}$. To verify our derived decay rate, we evaluate the general formula for a circular orbit ($e = 0 \implies K_e = 1$) with equal masses ($M_A = M_B = M$):

$$\begin{aligned} \dot{T} &= -\frac{192\pi}{5c^5}\left(\frac{2\pi G_N}{T}\right)^{\frac{5}{3}}\frac{M^2}{(2M)^{\frac{1}{3}}} = -\frac{192\pi}{5c^5}\frac{1}{2^{\frac{1}{3}}}\frac{M^{\frac{6}{3}}}{M^{\frac{1}{3}}}\left(\frac{2\pi G_N}{T}\right)^{\frac{5}{3}} \\ &= -\frac{192\pi}{5c^5}\frac{1}{2^{\frac{1}{3}}}\left(\frac{2\pi G_N M}{T}\right)^{\frac{5}{3}} = -\frac{192\pi}{5c^5}\frac{1}{2^{\frac{1}{3}}}\frac{1}{2^{\frac{5}{3}}}\left(\frac{4\pi G_N M}{T}\right)^{\frac{5}{3}} = -\frac{48\pi}{5c^5}\left(\frac{4\pi G_N M}{T}\right)^{\frac{5}{3}} \end{aligned}$$

Applying the full correction factor to the binary pulsar system PSR B1913+16 ($M_A = 1.441M_\odot$, $M_B = 1.378M_\odot$, $e = 0.617$, $T = 27900$ s) yields $\dot{T} \approx -2.4 \times 10^{-12}$ s/s. This matches observation to within 0.2%, constituting the first definitive evidence for gravitational radiation.